another gram of the same sample having been subjected in an open crucible and for not less than one hour to a temperature exceeding 925 degrees Celsius, and this total, plus the moisture previously determined, shall be recorded as incombustible matter.

Tests to be Recorded

6.44.13 The results of the tests of the roadway dust shall be recorded in a book kept at the mine for the purpose, and a copy posted at the mine entrance, and available on request.

Frequency of Tests

6.44.14 Tests of samples of roadway dust, so taken as to be representative of the normal composition of the roadway dust throughout the mine and on the floor, roof, and sides, respectively, shall be made as often as may be necessary, but not less frequently than once in each month.

Injurious Dust

6.44.16 Incombustible dust that is ineffective or injurious to health shall not be used in a mine.

Explosion Barriers – Coal Mines

Type

6.45.1 (1) In any underground coal mine which is dry and dusty, rock dust or water barriers of a type authorized by an inspector shall be installed at places designated in a scheme prepared by the manager and authorized by an inspector.

Inspection

(2) Where explosion barriers are required in an underground coal mine, the manager shall appoint a person who holds an underground coal mine shiftboss certificate to examine the condition and position of the barriers

Frequency

(3) The person appointed under subsection (2) shall examine the barriers at intervals of not more than 4 weeks and shall report the results of the examination in writing in a book to be kept at the mine.

Explosion Barriers – Other Mines

Other Mines

6.46.1 The chief inspector may order the manager to prepare a similar, authorized scheme at any mine where the chief inspector believes that a hazard may exist from the ignition of flammable gas or dust.

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Shafts

Shaft Design

7.1.1 (1) A mine shaft shall

(a) be of a design approved by a professional engineer,

(b) be equipped with a means to guide each shaft conveyance and counterweight through the shaft to prevent contact with another shaft conveyance or counterweight or with any shaft furnishing,

(c) have underwind clearances that exceed the stopping distance of any shaft conveyance when traveling at the maximum speed permitted by the hoist controls and carrying the maximum permitted load, except

(i) during shaft sinking operations, or

(ii) when chairs are used to land a skip during loading operations,

(d) where a friction hoist is installed, be equipped with tapered guides, or other devices approved by a professional engineer, located above and below the limits of regular travel of any shaft conveyance or counterweight, and designed to decelerate and safely stop the shaft conveyance or counterweight in the event of overtravel, and

(e) where a friction hoist is installed and where workers are transported in a conveyance not equipped with safety catches, safety chairs shall be installed in each compartment at the extreme limit of overwind travel; and the installation shall be so arranged so that if a conveyance or counterweights should break away from the rope as a result of an overwind, it would fall back the smallest practicable distance before landing on the safety chairs which, with their supports, shall be designed to stop and hold a fully loaded conveyance under these conditions.

Sump Barrier

(2) Except when a shaft is being sunk, or during sump cleaning operations, a barrier or obstruction shall be installed in the shaft to prevent a shaft conveyance from being lowered into water in the shaft bottom.

Guarding of Shaft Openings

7.1.2 A shaft opening in an underground mine shall be securely fenced, covered, or otherwise guarded.

Shaft Lining

7.1.3 (1) A shaft shall be securely lined and, if necessary, cased.

(2) During shaft sinking operations, the lining, and if necessary the casing, shall be maintained to within 20 m of the bottom of the shaft.

Shaft Compartments

7.1.4 (1) Except during shaft-sinking operations, a shaft compartment used for the handling of material shall be enclosed by a substantial partition at the collar and at all levels, except the side or sides on which material is loaded on or off the shaft conveyance.

(2) The enclosure referred to in subsection (1) shall

(a) extend above the collar and each level a distance of not more than 7 m, or less than the height of the shaft conveyance plus 2 m,

(b) extend below the collar and each level a distance of not less than 2 m, and

(c) conform to the size of the shaft conveyance, allowing for necessary operating clearances.

Shaft Passageways

7.1.5 (1) The passageway in a shaft shall be separated from the hoisting or counterweight compartments by a partition which complies with section 7.1.5 (2).

(2) Between levels, the partition may consist of metal or other material of suitable weight and mesh to prevent

(a) a falling object from entering the passageway, or

(b) the intrusion of an object from the passageway into the hoisting compartment.

(3) A safe passageway and standing room for a person outside the shaft shall be provided at all workings opening into a shaft and the passageway shall be directly connected with such openings.

Shaft Gates

7.1.6 (1) Except when the hoisting compartment at a shaft station is securely closed off, a substantial gate shall be installed.

(2) The gate required by subsection (1) shall be kept closed except when the shaft conveyance is being loaded or unloaded at the station, and have a minimum of clearance beneath it.

(3) The approach to the shaft shall be designed and protected to prevent inadvertent entry into the shaft of an uncontrolled vehicle.

Smooth Lined Shafts

7.1.7 (1) A passageway in a shaft shall be separated from the conveyance compartments by a mesh of suitable weight to prevent

(a) a falling object from entering the passageway, or

(b) the intrusion of an object from the passageway into the hoisting compartment.

Ladders in Shafts

7.1.8 (1) A suitable stairway or ladderway shall be maintained in every shaft.

(2) During shaft sinking operations, if the ladder is not maintained to the bottom of the shaft, an auxiliary ladder shall be provided.

(3) No ladder in a shaft, other than an auxiliary ladder used in shaft sinking operations, shall be installed in a vertical position.

(4) Where it is necessary for persons to perform inspections in a shaft, suitable ladderways, stairways, or platforms shall be provided to permit the work to be carried out in a safe manner.

Shaft Obstructions

7.1.9 (1) Protective devices and procedures shall be used to prevent a shaft conveyance from coming into contact with an intermediate shaft obstruction.

(2) A device which may become an intermediate shaft obstruction shall be positively locked out of the shaft compartment to prevent its inadvertent projection into the compartment.

(3) The location of each intermediate shaft obstruction shall be marked on the depth indicator of the hoist.

(4) The procedure for safely working with intermediate shaft obstructions shall be prepared in writing and posted in appropriate locations.

(5) Doors for covering the shaft at the collar, to facilitate the maintenance of a shaft conveyance, are not an intermediate shaft obstruction if

(a) they are positively latched out of the shaft compartments when not in use, and

(b) dual lights are installed to indicate to the hoist operator whether such doors are in or out of the shaft compartment.

Shaft Sinking

7.1.10 (1) When the vertical depth of a shaft below the collar exceeds 50 m during shaft sinking operations, including any preliminary underground development being carried out at the same time, a sinking bucket shall be provided for the transport of persons in the shaft.

(2) The sinking bucket shall be not less than 1.07 m high and designed in accordance with section 7.5.1 (1) and 7.5.1 (2).

(3) Where the distance between a head sheave and the shaft bottom exceeds 100 m, a crosshead shall be used with the bucket.

Sinking Crosshead

7.1.11 The crosshead shall

(1) land on at least two chairs at the bottom crosshead stop to prevent distortion,

(2) be attached to the rope by a safety appliance to ensure that the bucket is stopped if the crosshead should jam in the shaft, and

(3) be of a type that encloses the bucket, unless the shaft compartment is tightly lined and the bucket is barrel shaped.

Indicator Lights

7.1.12 Dual lights shall be installed within view of the hoist operator to indicate

(1) that the crosshead and the sinking bucket are descending the shaft together after leaving the dumping position, and

(2) whether the service doors or the dump doors are in or out of the shaft hoisting compartment.

Service Doors

7.1.13 During sinking operations, service doors shall be installed at the collar and at any other place in the shaft in order to cover the shaft compartments where the bucket is loaded or unloaded and the doors shall

(1) automatically latch out by mechanical means whenever removed from the hoisting compartment,

(2) be closed whenever a bucket is being loaded or unloaded with tools or other materials, and

(3) be closed whenever persons are entering or leaving the bucket, except where a closed crosshead is in use that provides equivalent protection for persons.

Dump Doors

7.1.14 During sinking operations, dump doors shall be installed at the bucket dumping position and arranged so as to:

(1) prevent a bucket from being dumped when the doors are in the open position,

(2) prevent any material from falling down the shaft when the bucket is being dumped, and

(3) automatically latch out by mechanical means whenever they are removed from the hoisting compartment.

Multi-deck Stages

7.1.15 Where a multi-deck stage is used for shaft sinking purposes, it shall be of a design approved by a professional engineer and constructed in accordance with that design.

Filling Bucket

7.1.16 A sinking bucket shall be filled so that no piece of loose rock projects above the level of its rim.

Riding in Bucket

7.1.17 Except during shaft maintenance and inspection operations, every person being transported by a bucket shall ride in the bucket when it is traveling above the bottom crosshead stopping point in the shaft.

Person in Charge

7.1.18 (1) The person authorized to give signals for hoist movement at a shaft sinking operation shall maintain proper discipline of persons riding in the bucket and enforce the permissible loading restrictions of the conveyance permit.

(2) No person shall obstruct the authorized person mentioned in subsection (1) from performing the authorized person's duties.

Steadying Bucket

7.1.19 A bucket shall not be allowed to leave the top or bottom of the shaft until it has been steadied.

Lowering Bucket

7.1.20 A bucket being lowered to the shaft bottom shall be

(1) stopped at a distance of at least 5 m, and not more than 10 m, above the bottom of the shaft, and

(2) lowered slowly beyond this point only on a separate signal.

Blasting Precautions

7.1.21 (1) During sinking operations, on the first trip carrying persons down the shaft after a blasting operation

(a) the bucket shall not be lowered beyond a point less than 15 m above the blasting set or bulkhead, or beyond any point in the shaft where the health or safety of the persons could be endangered, and

(b) only a sufficient number of persons shall be transported as required to make a proper examination of the parts of the shaft that might have been affected by the blast.

(2) Beyond the point specified in subsection (1), the bucket shall be lowered slowly on signals given by the persons riding in the bucket.

Overhead Protection

7.1.22 During shaft sinking operations

(1) persons may be at the bottom of the shaft during the hoisting and dumping cycles of the bucket, and

(2) no work shall be done in a shaft while persons are in another part of the shaft below such work unless there is a securely constructed covering over a sufficient portion of the shaft to protect the persons below from the danger of falling material.

Permits

Mine Hoist Certificates

7.2.1 (1) No hoist shall be put into service unless a certificate has been obtained from the manufacturer of the hoist or from a professional engineer, certifying
(a) the maximum rope pull,
(b) the maximum suspended load, and
(c) the maximum unbalanced load in the case of a friction hoist.

(2) No hoist shall be loaded beyond the maximum amount shown on the certificate required under subsection (1).

Mine Hoist Permit

7.2.2 Except for the purpose of testing before being put into service in a particular location, no mine hoist shall be operated without a letter of certification issued by a professional engineer, a copy of which shall be posted in the hoistroom and provided to an inspector.

Shaft Conveyance Permit

7.2.3 (1) No shaft conveyance shall be used without a valid shaft conveyance permit.

(2) A shaft conveyance Letter of Certification shall be obtained from a professional engineer and a copy posted in the hoistroom.

(3) A certificate shall be obtained for each shaft conveyance and counterweight showing its
(a) rated load, as certified by a professional engineer, and
(b) serial number, date of manufacture, and the name of the manufacturer.

(4) Each shaft conveyance and counterweight shall be recertified in accordance with subsection (3) every 5 years.

Commissioning Tests

7.2.4 (1) Commissioning tests shall be conducted on a new or reactivated mine hoisting plant to ensure compliance with the code.

(2) The results of such tests shall be recorded and a copy sent to an inspector.

(3) The person responsible for commissioning the mine hoisting plant shall be a professional engineer who shall certify the results of the commissioning tests.

(4) No mine hoisting plant shall be placed into operation prior to the successful completion of commissioning tests.

Maximum Number of Persons in a Conveyance

7.2.5 (1) Subject to subsection (2), the maximum number of persons that may be carried in a shaft conveyance shall be determined as follows

(a) where the clear floor area of the deck of a shaft conveyance is 1.86 square metres or less, there shall be at least 0.19 square metre for each person,

(b) where the clear floor area of a deck of a shaft conveyance is more than 1.86 square metres and less than 4.64 square metres, there shall be at least 0.16 square metre for each person, and

(c) where the clear floor area of a deck of a shaft conveyance is 4.64 square metres or more, there shall be at least 0.14 square metre for each person.

(2) The maximum number of persons that may be carried by a shaft conveyance shall not exceed 85 per cent of the maximum allowable weight of materials divided by 90 kg.

(3) No shaft conveyance shall be loaded with more than the maximum number of persons or weight of material specified in the shaft conveyance permit and a notice showing the specified limits shall be posted at the shaft collar.

Logbooks

7.2.6 The following logbooks shall be obtained and used for each mine hoisting plant

(1) Electrical Hoisting Equipment Record Book,

(2) Hoisting Machinery Record Book,

(3) Hoist Operator's Logbook,

(4) Rope Record Book, and

(5) Shaft Inspection Record Book.

Hoist Operator's Certificate

7.2.7 (1) The operator

(a) is the holder of a current certificate from a medical practitioner stating that the operator has been examined, and is fit to discharge the duties of a hoist operator, and

(b) is qualified and authorized to operate the hoist or, in the case of a worker being trained as a hoist operator, is under the direct supervision of a qualified and authorized person.

(2) The medical certificate shall expire one year after its date of issue.

(3) Notwithstanding subsection (2), a hoist operator who has suffered from an accident or illness that, in the opinion of the mine manager, might impair the hoist operator's efficiency shall obtain a medical certificate before returning to work.

(4) A record of all medical certificates shall be kept on file.

Headframes and Sheaves

Headframe Design

7.3.1 A shaft headframe, whether on the surface or underground, shall

(1) have the plans of the design certified by a professional engineer, and

(2) be of sufficient height to provide an overwind distance that exceeds the greater of twice the stopping distance of the shaft conveyance traveling at the maximum speed permitted by the hoist controls, or 3 m.

Sheave Certificate

7.3.2 (1) Before a shaft rope sheave is used, a certificate shall be obtained from the manufacturer or from a professional engineer certifying,

(a) its maximum rated load carrying capacity,

(b) the diameter of rope for which it was designed,

(c) the maximum breaking strength of the rope for which it was designed, and

(d) the maximum amount of groove wear that shall be permitted.

(2) No shaft rope sheave shall be loaded beyond its maximum rated load carrying capacity or used otherwise than in compliance with the certificate required by subsection (1).

Rope and Sheave Grooves

7.3.3 The ratio of the diameter of the sheave groove to the diameter of the shaft rope shall be as required by section 7.6.1.

Sheave Construction

7.3.4 A shaft rope sheave shall

(1) be made of material that can safely withstand the ambient temperatures where it is installed,

(2) have a groove, or inserts with a groove, which is suitable for the rope being used, and

(3) bear a serial number and the date of its manufacture.

Sheave Shaft

7.3.5 The shaft of a sheave shall be non-destructively tested for flaws by a certified person before being put into service in a particular location, and after installation in accordance with section 7.9.7.

Ropes

Rope Attachments

7.4.1 (1) Each component of the suspension gear between a shaft conveyance or counterweight and a hoisting or tail rope, and any connections between shaft conveyances, shall have a static factor of safety of not less than 10.

(2) Components of a single suspension unit between a hoisting rope and a shaft conveyance shall not be welded.

(3) A shaft rope shall be attached to the suspension gear of a shaft conveyance or counterweight by a closed type device that cannot be inadvertently disconnected.

Rope to Drum Attachment

7.4.2 (1) In a drum hoist installation, the hoisting rope of a shaft conveyance or counterweight shall be attached to the drum of the hoist.

(2) There shall not be less than three rounds of rope on the drum of a drum hoist when the shaft conveyance is at the lowest point in the shaft from which hoisting is possible.

Trial Trips

7.4.3 When the attachments for a shaft hoisting rope are first installed, or reinstalled after dismantling, the following procedures shall be performed before the hoist is put into service

(1) two test trips of the conveyance or counterweight shall be made through the working part of the shaft, with the conveyance carrying its normal maximum load,

(2) the attachments shall be examined by a qualified person upon completion of the two test trips and any necessary adjustments made, and

(3) a record of the test trips, and any adjustments or examinations made, shall be entered in the Hoisting Machinery Record Book by the person or persons making the tests, examinations or adjustments.

Rope Test Certificate

7.4.4 (1) Before a shaft rope is initially installed,

(a) a 2.5 m representative sample shall be cut from the rope and sent for destructive testing to an acceptable testing laboratory

(b) a shaft rope that has been in storage for more than 12 months shall be retested in accordance with subsection (a).

(c) a test certificate is issued by the laboratory showing the actual breaking strength of the rope, the percentage elongation at failure and the results of torsion tests performed on individual wires.

(2) When the rope is installed, a copy of the Test Certificate is sent to the inspector with the appropriate pages from the Rope Record Book.

Rope Destructive Test

7.4.5 After a shaft rope has been in service for 6 months on a drum hoist, and at subsequent intervals not exceeding 6 months, the manager shall ensure that

(1) a 2.5 m length shall be cut off from the lower end of the rope above the clamps, the cut ends securely bound and the sample sent for a destructive test at an acceptable rope testing laboratory.

(2) a test certificate showing the information required by section 7.4.4 (1) (c) shall be obtained from the laboratory and forwarded without delay to the inspector.

Rope Electromagnetic Test

7.4.6 A qualified person, shall test each shaft rope throughout its working length with an electromagnetic rope testing device.

(1) within 6 months of being put into service and at subsequent intervals of not longer than 6 months, or

(2) at intervals shorter than 6 months if an extrapolation of previous test results indicates that the loss of breaking strength of the rope could exceed 10% before the next regular test, or

(3) at intervals of not longer than 4 months when the rope has been in service for more than 2 years.

Records

7.4.7 The date of every electromagnetic test carried out and the results obtained shall be entered in the Rope Record Book required by section 7.2.6.

Copy for the Inspector

7.4.8 As record of every electromagnetic test performed, including graphs and their interpretation, signed by the person making the interpretation, shall be sent to the inspector within 28 days of the completion of the test or, if the test indicates a loss of strength of the rope in excess of 7.5%, within 14 days.

Minimum Rope Diameters

7.4.9 The minimum nominal diameter of a hoisting rope shall be

(1) 16 mm (5/8") where a shaft conveyance or counterweight is supported by a single rope, and

(2) 13 mm (1/2") where a shaft conveyance or counterweight is supported by more than one rope.

Factors of Safety Drum Hoist Ropes

7.4.10 The factor of safety of a hoisting rope installed on a drum hoist shall not be less than

(1) 8.5 at the point where the rope is attached to the shaft conveyance when carrying its maximum permissible load, or to the counterweight,

(2) 5.0 at the point where the rope leaves the head sheave when the shaft conveyance is carrying its maximum permissible load or, the counterweight is at its lowest point of normal travel, and

(3) 7.5 at the point where the rope is attached to a skip or counterweight if the maximum material load being carried is accurately determined by weighing.

Factor of Safety Friction Hoist Ropes

7.4.11 The factor of safety of a hoisting rope installed on a friction hoist shall not be less than the greater of the value obtained from the formula 8.0 minus 0.00 164 d, where d is the maximum length of rope, in meters, in the shaft compartment below the head sheave or the friction hoist drum, or 5.5.

Factor of Safety Balance Ropes

7.4.12 The factor of safety of a tail or balance rope shall not be less than 7.0.

Shaft Rope Record

7.4.13 Whenever a shaft rope is installed or removed from service the appropriate pages in the Rope Record Book required by section 7.2.6, shall be fully completed and a copy sent to the inspector.

Rope Not to Be Used

7.4.14 No rope shall remain in service as a shaft rope when its breaking strength is found to be less than the following percentage of the breaking strength shown on the pre-installation test certificate required by section 7.4.4(2)

(1) in any part of a hoisting rope, 90%,

(2) in any part of a multi-layer, multi strand balance rope, 90%, and

(3) in any part of a single layer, stranded balance rope, 85%.

Rope to Be Removed

7.4.15 Notwithstanding section 7.4.14, no rope shall remain in service as a shaft rope if

(1) the extension of a test piece of the rope has decreased to less than 60% of the original extension shown on the test certificate required by section 7.4.4(2),

(2) the number of broken wires, excluding filler wires, in any section of the rope equal to one lay length, exceeds 5% of the total number of wires in the rope, excluding filler wires, or

(3) the rate of stretch in a hoisting rope, installed on a friction hoist, shows a rapid increase in the normal rate of stretch recorded during its service.

Reporting Defects to Manager

7.4.16 (1) If, during any examination or test required to be carried out by Part 7 of the code a weakness or defect is discovered which could endanger the safety of any person, the matter shall be reported immediately to the manager and the hoisting equipment shall not be used until the necessary remedial action has been taken.

(2) If any examination or test reveals an unusual condition of a shaft rope, although not constituting sufficient reason for condemning the rope a more frequent level of examination shall be conducted until the rope is certified by a professional engineer.

(3) The unusual condition and results of the inspection shall be recorded in the Shaft Rope Record Book.

Conveyances

7.5.1 (1) All parts of a shaft conveyance or counterweight, when carrying the rated load, shall be capable of withstanding at least 4 times the maximum allowable design stress without permanent distortion.

(2) The maximum allowable design stress shall be established by good engineering Practice and include the effects of

(a) the weight of the conveyance or counterweight,
(b) the rated load,
(c) any impact load,
(d) any dynamic load,
(e) stress concentration factors,
(f) corrosion,
(g) metal fatigue, and
(h) dissimilar materials.

Worker Protection

7.5.2 Where a person performs work from the top of a shaft conveyance or counterweight, the manager shall ensure that there is provided a safe footing, and except when changing shaft guides, protection from falling objects.

Transport of Supplies and Equipment

7.5.3 (1) Shaft conveyances shall be equipped with devices for safely securing any equipment or supplies that are to be transported in the shaft.

(2) Whenever equipment or supplies are being transported in a shaft, they shall be loaded and secured so as to prevent them from shifting.

(3) When any material being transported in a shaft projects above the top of the shaft conveyance, the projecting portion shall be securely fastened to the shaft conveyance and not to the hoist rope.

(4) If equipment is slung underneath a conveyance or crosshead to be transported in a shaft, it shall be suspended in a manner to prevent its contact with any part of the shaft, its lining, or its furnishing.

(5) All components of a suspension system used to sling equipment underneath a conveyance or crosshead being transported in a shaft shall be capable of withstanding, without permanent deformation, at least 4 times the maximum allowable design stress determined in accordance with section 7.5.1 (2).

Transporting Persons

7.5.4 Except during shaft sinking operations, when a shaft exceeds 60 m in vertical depth a suitable shaft conveyance shall be provided for the transport of persons in the shaft.

Cage Requirements

7.5.5 A cage that is used for the transport of persons in a shaft shall

(1) where it is supported by a single rope or a single point of attachment, be equipped with safety catches and mechanisms as prescribed in section 7.5.10,

(2) except on any side with a door, be enclosed by steel sheet at least 3 mm thick or by material of equivalent strength,

(3) have a hood made of steel plate, at least 5 mm thick or of material of equivalent strength,

(4) have an internal height greater than 2.1 m,

(5) have a clearance at the door greater than 1.8 m, and

(6) where practical, have an exit in the roof, which can be opened from inside or outside the cage, through which persons can pass.

Cage Doors

7.5.6 The doors on a cage that is used to transport persons in a shaft shall

(1) be at least 1.5 m in height,

(2) be mounted and arranged so that they cannot be opened outward from the cage,

(3) be of solid construction except for a viewing window,

(4) be arranged so that they can be closed whenever persons or materials, other than rolling stock, are being transported in the cage,

(5) be installed so that there is only enough clearance at the floor to allow them to be readily opened and closed, and

(6) be of adequate strength to withstand normal shock loads.

Transport of Persons in Skip

7.5.7 Except when being used, to transport workers for shaft inspection, maintenance purposes, or when an emergency situation occurs, a skip shall not be used to transport persons.

(1) When a skip is being used to carry persons, the hoist shall
   (a) be equipped with control devices that prevent the skip from being taken to the dump position and to the skip loading pocket, unless the controls for loading the skip with ore or waste have been locked out in accordance with section 4.11.2,
   (b) not be permitted to travel in excess of one-half its normal speed and in no case shall the speed be permitted to exceed 5 m/s, and
   (c) a skip shall have a suitable platform provided for transporting persons so as to prevent any part of the body protruding outside.

(2) The control devices of the hoist shall be designed and installed to fail to safety.

(3) An audible or visible signal that the control devices for the hoist are set in operation shall be given to persons entering a skip.

Fall Protection

7.5.8 The opening between the side of a shaft and the skip box over which a person must pass when entering or leaving the skip shall be closed off sufficiently to prevent a person from falling through the opening.

Signal Pull Cord

7.5.9 The shaft signal pull cord shall be located in a convenient place for the skip tender when persons are being transported in the conveyance.

Safety Catches

7.5.10 The safety catches on a cage or skip shall

(1) be of a type and design approved by a professional engineer,

(2) safely stop and hold a cage or skip carrying persons if the hoisting rope or its attachment should fail, and

(3) be subjected to, and successfully pass, free fall tests as prescribed in sections 7.5.11 and 7.5.12
   (a) before the cage or skip is first used for the transport of persons, and
   (b) before the cage or skip is used for the transport of persons after repairs to correct any defect or distortion of the safety catches and mechanisms.

Free Fall Tests

7.5.11 The free fall tests required by section 7.5.10 (3) shall be performed under the following conditions

(1) the cage or skip shall carry a weight equal to the maximum permitted load of persons carried and of any material permitted to be carried at the same time,

(2) the cage or skip shall travel at a speed equal to the normal hoisting speed when transporting persons, and

(3) the guides on which the test is made shall be representative of those in the shaft.

Free Fall Tests – Results

7.5.12 A free fall test shall be considered successful if

(1) the cage or skip is decelerated within 1 g and 3 g,

(2) there is no damage to the safety dogs or mechanisms,

(3) the safety dogs engage the guides continuously during deceleration, and

(4) calculations show that the safety dogs would stop the cage or skip when carrying its maximum permitted material load.

Free Fall Tests – Report

7.5.13 The results of a free fall test shall be submitted to an inspector and entered in the Hoisting Machinery Record Book.

Chairs

7.5.14 Chairs used for landing a cage shall be

(1) arranged to fall clear and remain clear of the shaft compartment when the cage is lifted off the chair,

(2) operable only from outside the cage, and

(3) so arranged as not to distort the cage.

Work Platforms in Shafts

7.5.15 (1) Where a work platform that is not a shaft conveyance is used to transport or support a person, it shall be designed by a professional engineer and built in accordance with the design.

(2) The manager shall prepare a written procedure for the use of a work platform in a shaft before putting it into service.

Hoists

Drum to Rope Ratio

7.6.1 (1) The ratio of a hoist drum diameter to the rope diameter shall be equal to or greater than

(a) in the case of a drum hoist, 60 to 1 where the nominal rope diameter is 26 mm (1") or less, and 80 to 1 where the nominal rope diameter is greater than 26 mm (1"),

(b) in the case of a hoist used for shaft sinking or for preliminary development work during shaft sinking, 48 to 1 where the nominal rope diameter is 26 mm (1") or less, and 60 to 1 where the nominal rope diameter is greater than 26 mm (1"), and

(c) in the case of a friction hoist, 80 to 1 for stranded ropes, and 100 to 1 for locked coil ropes.

(2) Subsection (1) (a) does not apply to a drum hoist installed and approved for use prior to the code coming into force.

Hoist Brakes Required

7.6.2 (1) No hoist shall be used for the transport of persons in a shaft unless it has at least two sets of mechanical brakes to stop and hold the drum from which persons are being hoisted or lowered,

(2) Each set of mechanical brakes shall

(a) be capable of safely stopping and holding the drum when the shaft conveyance is carrying its maximum permitted load,

(b) be arranged so that it can be independently tested,

(c) be arranged to apply normal braking effort before any linkage or brake piston reaches a limit of travel, and

(d) if of a drum type, be equipped with a device to give positive indication of tread wear or slack linkage, and prevent any movement of the hoist if predetermined limits are exceeded.

Brake Application

7.6.3 At least one set of mechanical brakes required by section 7.6.2 shall be designed and arranged to

(1) apply directly to the hoist drum, and

(2) apply automatically whenever the safety circuit of the hoist is interrupted and whenever the pressure in the hydraulic or pneumatic brake actuating system drops below normal.

Brake Control

7.6.4 The mechanical braking system of a hoist shall be arranged so that

(1) the brake is applied by a control lever that is pulled, unless there is a common brake and power lever,

(2) where brake weights are installed to provide braking force, they can be readily tested to ensure freedom of movement, and

(3) at least one set of hoist brakes is applied automatically if there is a loss of power to the hoist.

Hoist Drum Deceleration

7.6.5 (1) Subject to subsection (2), the brakes of a drum hoist shall be designed and arranged to decelerate the drum at between 1.5 and 3.7 m/s/s

when braking is initiated by an interrupted safety circuit and the hoist is

(a) normally used for transporting persons, and

(b) operating in the normal full speed zone in the shaft.

(2) Subsection (1) does not apply to a drum hoist installed and approved for use prior to the code coming into force.

Brake Testing Required

7.6.6 The braking system of a hoist not normally used for the transport of persons shall be designed and arranged to safely stop and hold the shaft conveyance under all permitted conditions of loading and speed and in either direction of travel.

Foot-operated Brake

7.6.7 No hoist shall be equipped with a foot-operated brake.

Protective Circuits and Hoist Safety Devices

7.6.8 (1) No electric hoist, regardless of operation, shall be used unless it has a fail-safe, protective electrical circuit at a normal potential not exceeding 250 V and all safety circuits of the electric hoist shall be installed in accordance with CSA M421-00 Use of Electricity in Mines.

(2) The track limit switch shall be installed in each shaft hoisting compartment above the normal upper limit of travel, and so arranged and positioned that in the event of an overwind it will be operated directly by the shaft conveyance or counterweight to interrupt the hoist safety circuit and bring the hoist to a safe stop before the conveyance or counterweight or rope attachments can reach any permanent obstruction in the shaft or headframe.

Synchronization

7.6.9 A device shall be installed on each friction hoist to synchronize the position of the shaft conveyance with the safety devices driven from the hoist drum.

Electric Hoist – Requirement

7.6.10 Every electric hoist shall have a speed indicator if the normal rope speed exceeds 2.5 m/s and a device which gives a voltage reading proportional to the speed of the hoist.

Electric Hoists – Safety Devices

7.6.11 All safety circuits of an electric hoist shall be installed in accordance with CSA M421-00 Use of Electricity in Mines.

Electric Hoists – Controls

7.6.12 Every electric hoist shall have

(1) a master controller that has a neutral or brake reset position,

(2) the brake operating levers arranged so that, after any interruption of the hoist safety circuit, power cannot be restored to the hoist unless the brake levers are in the brake-applied position, and

(3) accurate and sensitive safety controllers.

Safety Devices – Design and Adjustment

7.6.13 Every safety and protective device installed at a hoisting installation shall

(1) be designed and installed taking into account the environmental conditions at its operating location, and only be adjusted and maintained by a qualified and authorized person.

Indicators

7.6.14 A hoist shall be provided with depth indicators that continuously, accurately, and clearly show the hoist operator the position

(1) of a shaft conveyance and counterweight, if any,

(2) in an inclined shaft, of a change in gradient that requires a reduction in hoist speed,

(3) at which the overwind, underwind, and track limit devices are set to operate,

(4) of any intermediate shaft obstruction,

(5) of the limits of normal travel for the shaft conveyance and counterweight, if any, and

(6) of any collar doors, dump doors, and crosshead landing chairs.

Drum Requirements

7.6.15 No drum hoist shall have

(1) more than 3 layers of rope where the drum has helical or spiral grooving or does not have grooving,

(2) more than 4 layers of rope if the drum has parallel and half pitch grooving, and

(3) less than 3 dead turns of rope remaining on the drum when the conveyance or counterweight is at its lowest possible position in the shaft.

Drum Grooves and Flanges

7.6.16 (1) A cylindrical drum on a hoist shall be provided with

(a) grooves that properly fit the rope in use, except that during shaft sinking operations and preliminary development work, a smooth drum may be used, and

(b) flanges of sufficient height to contain all of the rope on the drum and of sufficient strength to withstand any loads imposed by the rope.

(2) Any conical portion of a hoist drum shall be provided with grooves to prevent the rope from slipping on the drum.

(3) A drum hoist and a sheave shall be arranged so that the rope coils properly across the face of the drum and winds smoothly from one layer to another without cutting into the rope layer beneath.

Brake/Clutch Interlock

7.6.17 (1) A clutch of a drum hoist shall be interlocked with the brake so that
(a) the clutch can be disengaged only when the brake of the drum is fully applied, and
(b) the clutch is fully engaged before the drum can be released.

(2) The controls for engaging and disengaging a clutch shall be designed or equipped with guards to prevent their inadvertent operation.

(3) A band type friction clutch shall not be used.

Slack Rope Detection

7.6.18 Drum hoists shall be equipped with a device for detecting slack hoist rope.

Communications and Signaling

Cage Call System

7.7.1 Cage call systems shall be installed so that the signals are audible to the hoist operator and electrical components are installed as per CSA M421 Use of Electricity in Mines.

Hoist Signaling System

7.7.2 A signaling system shall be installed at every hoisting shaft to permit the person in charge of the shaft conveyance and the hoist operator to exchange hoist control signals.

Signaling System – Electrical Requirements

7.7.3 (1) The signaling system required by section 7.7.2 shall be supplied with power at a potential not exceeding 150 V from a transformer which does not supply any other equipment.

(2) If the primary voltage of the transformer exceeds 750 V
(a) one conductor of the power supply shall be grounded, or
(b) if the conductors are ungrounded, an isolating transformer with a one-to-one ratio shall supply power for the signaling system, and
(c) a device shall be installed to indicate a ground fault.

(3) All metal parts of the signaling system which do not carry electrical current shall be grounded unless located not less than 2.4 m above the floor.

Multi-deck Stage Signals

7.7.4 A shaft signaling system installed on a multi-deck stage at a sinking operation shall be capable of providing a signal that is both audible and visible.

Radio Frequency Signals

7.7.5 A shaft signaling system using radio frequencies shall

(1) be tested to determine if there is any hazard to the use of blasting caps in the mine, and

(2) only be operated if precautions are taken to prevent the risk of an inadvertent or accidental detonation of any explosive material as a result of radiated energy.

Persons Authorized to Give Signals

7.7.6 (1) Only qualified persons authorized by the manager shall give any shaft signal other than the danger signal.

(2) Except during maintenance work, shaft sinking, or preliminary development work, or to recall a conveyance after a 5 bell release, no person shall give any shaft signal unless the shaft conveyance is located at the level from which the signal is being given.

(3) No person shall interfere in any way with the signaling arrangements.

Signaling System Operation

7.7.7 (1) A hoist shall not be moved on manual control unless

(a) the signal prescribed in section 7.7.9 has been given and returned by the hoist operator,

(b) at least 4 seconds have elapsed after the executive signal has been given, and

(c) if the hoist operator is unable to act within 30 seconds of receiving the executive signal, until a complete signal has again been received.

(2) Shaft signals shall be given in the following sequence

(a) cautionary,

(b) destination, and

(c) executive.

7.7.8 The prescribed code of signals set out in Appendix 7-1 of this part shall be used at every mine, and a copy of the signal code posted in every hoistroom, at the collar of the shaft, and at every working level or other landing in the shaft.

Special Signals

7.7.9 (1) Special signals in addition to , and readily distinguishable from, those prescribed in section 7.7.8 shall
(a) be used to designate hoist movements without interfering in any way with the prescribed signals, and
(b) be approved by the inspector.
(2) The special code of signals in use at a mine and an adequate description of their application shall be posted in the hoistroom, at the shaft collar, and at every working level of the shaft.

Voice Communications

7.7.10 A system for communicating by voice shall be installed and maintained to permit communication between persons at the collar of the shaft, the landing stations, and the hoistroom for the shaft.

Hoisting Procedure

Transport of Persons

7.8.1 Persons shall not be transported in a shaft conveyance that is simultaneously being used to transport ore, or other material, or when the cage doors are open.

Transport of Persons With Supplies

7.8.2 Where a multi-deck cage is in use, no persons shall be allowed to travel in the cage when it is loaded with supplies or rolling stock, except that persons may be carried on the top deck with the doors closed if
(1) the materials and rolling stock are loaded on another deck and adequately secured, or
(2) the combined load does not exceed 85% of the material load limit of the cage.

Transport of Persons With Personal Equipment

7.8.3 No persons shall be allowed to travel in a shaft conveyance when personal hand tools or equipment are being transported unless
(1) they are adequately secured and guarded where necessary, and
(2) the combined load does not exceed 85% of the material load limit of the conveyance.

Authorized Person in Charge of Conveyance

7.8.4 (1) Whenever persons are being transported in a shaft conveyance where the hoist is operated under manual control, there shall be a qualified and authorized person in charge of the conveyance at all times who shall